IoT Systems Activity 1

IoT Messaging Platform Activity

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Part 1: Basic Publisher and Subscriber

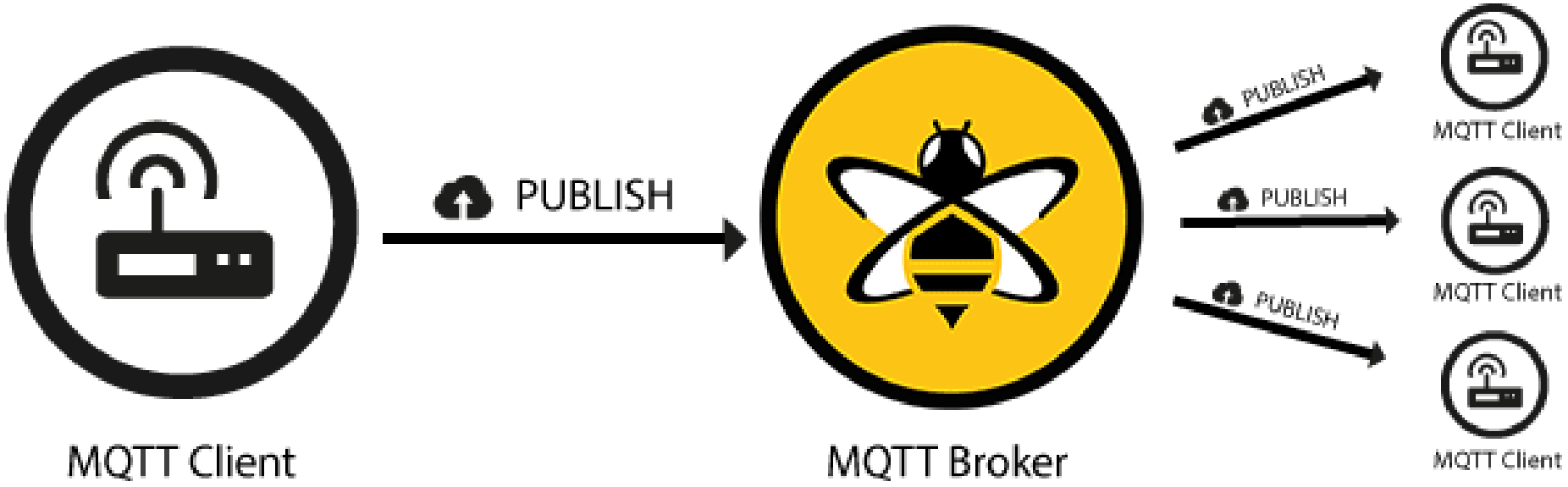


Image from: https://www.hivemq.com/blog/mqtt-essentials-part-4-mqtt-publish-subscribe-unsubscribe/

Activity Part 1 Brief

1. Implement a publisher and a subscriber in *any* programming language of your choice.

2. Connect to the provided broker

BROKER_HOST = 161.200.92.6
BROKER_PORT = 27004

3. Publish messages (payload) to the broker

Topic: **IotMsgPlatformPartOne/{YourGroupName}**

```
{
  "mac_address": "AA:BB:CC:DD:EE:FF"
}
```

Activity Part 1 Brief

4. Subscribe to the topic **“IotMsgPlatformPartOne/respond”**. You will receive the following payload

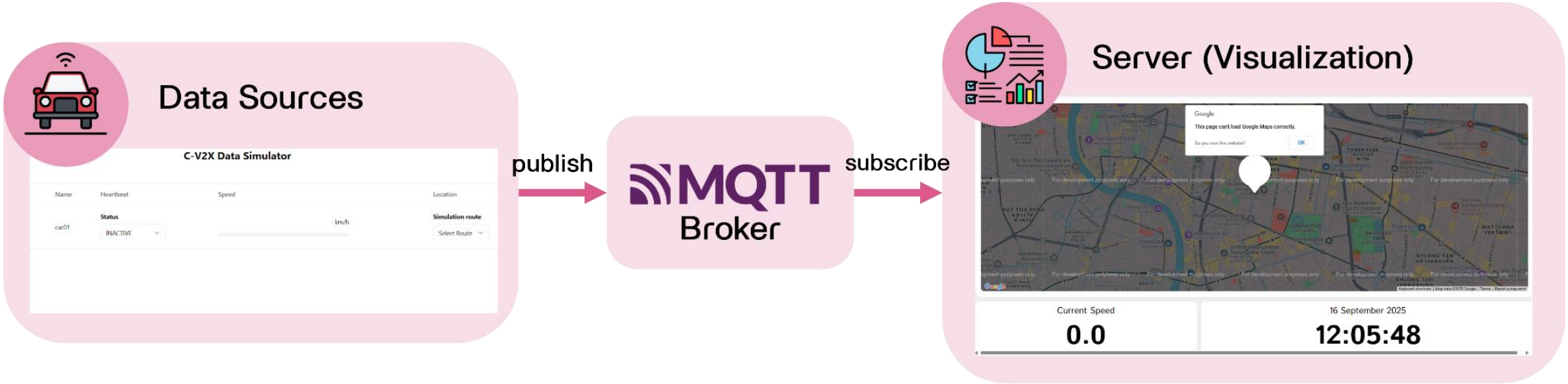
Topic: **IotMsgPlatformPartOne/respond**

```
{
  "group_id": "IoT",
  "v2x_topic": "IotTopicPartTwo",
  "error": ""
}
```

5. Save the received **v2x_topic**. You will need it in the part 2

6. Answer the Part 1’s questions in myCourseVille

Part 2: IoT Messaging Platform in Internet of Vehicles (IoV)



Activity Part 2 Brief

1. Clone the git repository

GITHUB_URL = https://github.com/Ricecamera/mqttActivity_IoTCEDT.git

2. Follow the instructions in the activity document to install dependencies and start the website. If you have done it correctly, you should be able to open these webpages

Vehicle Dashboard: http://localhost:3000/dashboard

C-V2X Data Simulator: http://localhost:5173

Activity Part 2 Brief

3. Implement MQTT publisher in **Sensor/server/obu.js**. Send payload to the following topics:

- <v2x_topic>/speed
- <v2x_topic>/heartbeat
- <v2x_topic>/route

4. Implement MQTT Subscriber in **OBU/consumer.js**

5. After completing both publisher & subscriber, contact TAs to review and grade your work.

6. Answer the Part 2's questions in myCourseVille

You may need to deploy a broker yourself to test the implemented publisher and subscriber.